



TRUE SITE

**AI-Powered Spoofing Detection And
Source Identification**



TEAM MEMBER

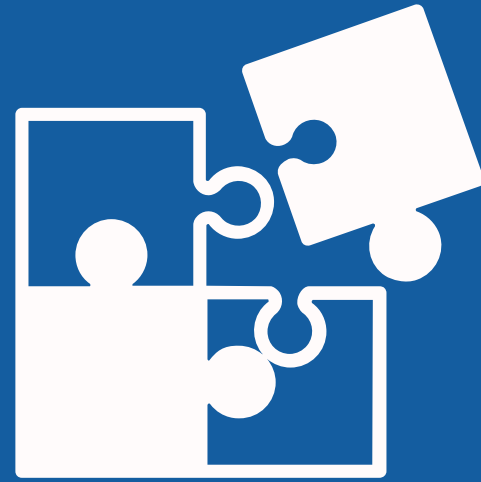
- Asad Ali (2022903)
- Ammaid Saleem (2022344)
- Humayun Nasir (2022189)
- Aizaz (2022040)

Supervisor

Dr. Zain Zidique

Co - Supervisor

Dr. Khurram Jaddon



Evaluation Criteria

- ✓ Problem statement clearly defined
- ✓ Project overview explained
- ✓ Motivation and ethical impact justified
- ✓ Literature review with comparison
- ✓ In-depth engineering/computing knowledge

Project Overview

Key Features

- Real-Time Multi-Modal Authentication
- Lip-Sync Discrepancy Detection
- Facial Liveness Verification
- Urdu-Language Audio Analysis
- Source Identification via Metadata
- Behavioral Anomaly Detection

Ethical Impact

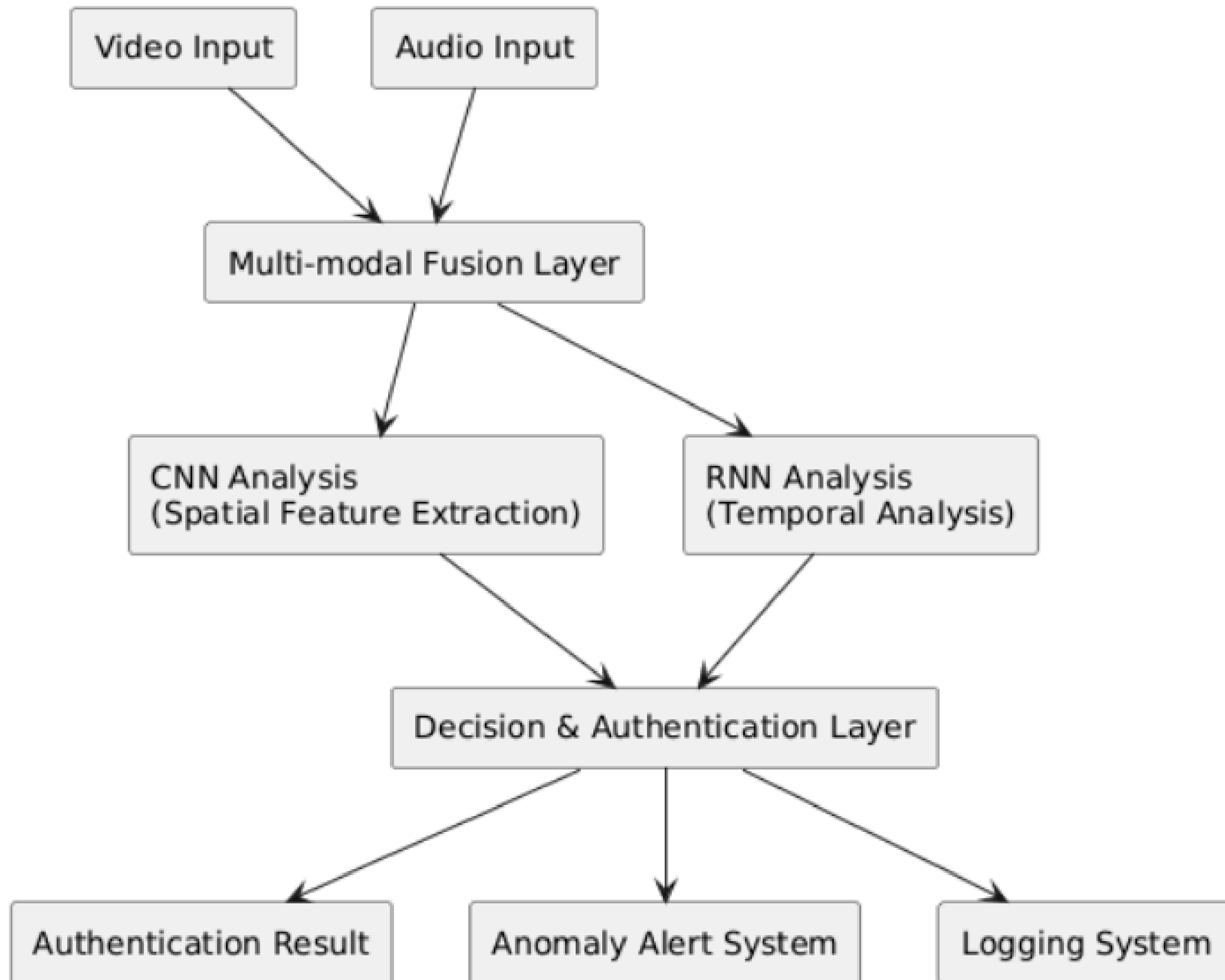
- Prevents Identity Theft
- Supports Digital Trust
- Promotes Responsible AI Use



Problem Statement

- **Core Issue:** The sophistication of generative AI (GANs, Diffusion Models) has rendered traditional unimodal authentication (face or voice) vulnerable to advanced presentation attacks.
- **Critical Gap:** Existing solutions lack robust forensic source identification, making accountability and traceability impossible.
- **Domain-Specific Challenge:** A significant lack of detection models tailored for low-resource languages like Urdu.
- **Impact:** This poses a direct threat to the security of real-time communication platforms in digital banking, legal proceedings, and remote identity verification.

AI Processing Engine

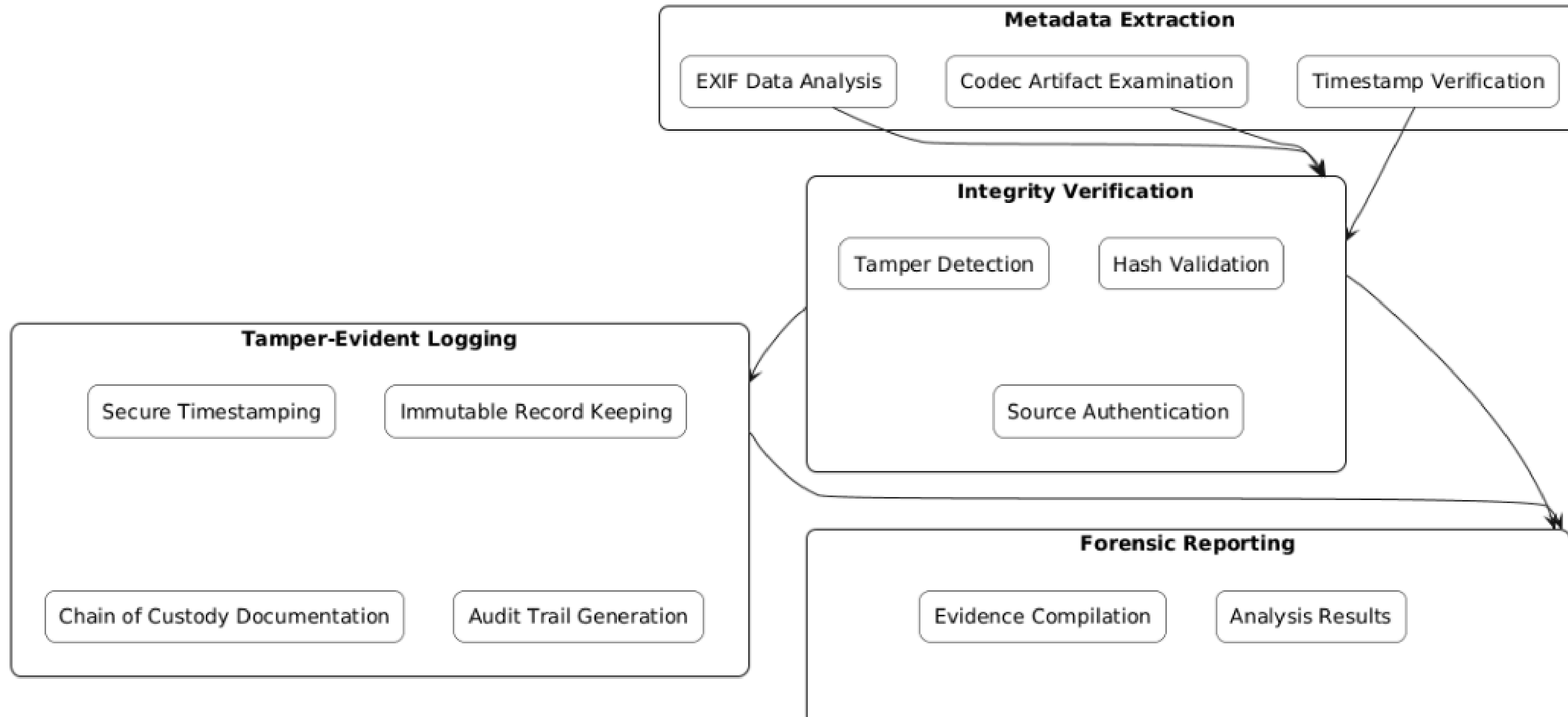


Technical Approach

FORENSIC COMPONENTS



Forensic Analysis Engine





Literature Review – Comparison with XceptionNet

Feature / Limitation	XceptionNet (Existing Model)	Truesite (Proposed)
Detection Scope	Video frame-based (image-level dee detection)	Multi-modal (audio 🎤, video 📹, text 📄, images 🖼️)
Language Support	Works only on English/visual dataset	Includes Urdu + multilingual
Source Identification	✗ Not supported	✓ Full forensic source tracing
Attribution Accuracy	Limited to face forgery classification	High (metadata + behavioral + ML fusion)
Deployment	Research-grade, not real-time	Real-time scalable system



Innovation & Differentiation

1

Combined audio-video-text analysis

2

Urdu-language specific detection

3

Real-time source identification

4

Behavioral pattern recognition

5

Adaptive threat response

Implementation Plan

- Phase 1: Data Collection & Preprocessing
- Phase 2: Model Development & Training
- Phase 3: System Integration
- Phase 4: Testing & Validation
- Phase 5: Deployment Preparation



Implementation Plan

Phase 1: Data Collection & Preprocessing



Datasets: Utilize SiW, ASVspoof; create novel Urdu audio dataset.

Synthetic Data: Generate spoofs using GANs (StyleGAN2) & diffusion models for augmentation.

Processing: Face alignment (MTCNN), audio conversion to Mel-spectrograms (Librosa).

Phase 2: Model Development & Training



Liveness Detection: CNN (EfficientNet) with pseudo-depth map supervision.

Temporal Analysis: RNN (LSTM) for lip-sync and sequence validation.

Audio Deepfake Detection: CNN & LightGBM on Mel-spectrograms for Urdu.

Implementation Plan



Phase 3: System Integration

- Fusion: Feature-level fusion of video, audio, and temporal model outputs into a Dense network.
- API: Build a real-time processing pipeline with a FastAPI backend.⁴

Phase 4: Testing & Validation

- Metrics: ACER, F1-Score, Latency.
- Methods: k-fold cross-validation, adversarial testing with deepfake datasets (FaceForensics++).

Phase 5: Deployment Preparation

- Containerization: Docker for environment consistency.
- Optimization: Model quantization & pruning for faster inference.

Expected Outcomes

Deliverables:

Real-time Deepfake Detection System:

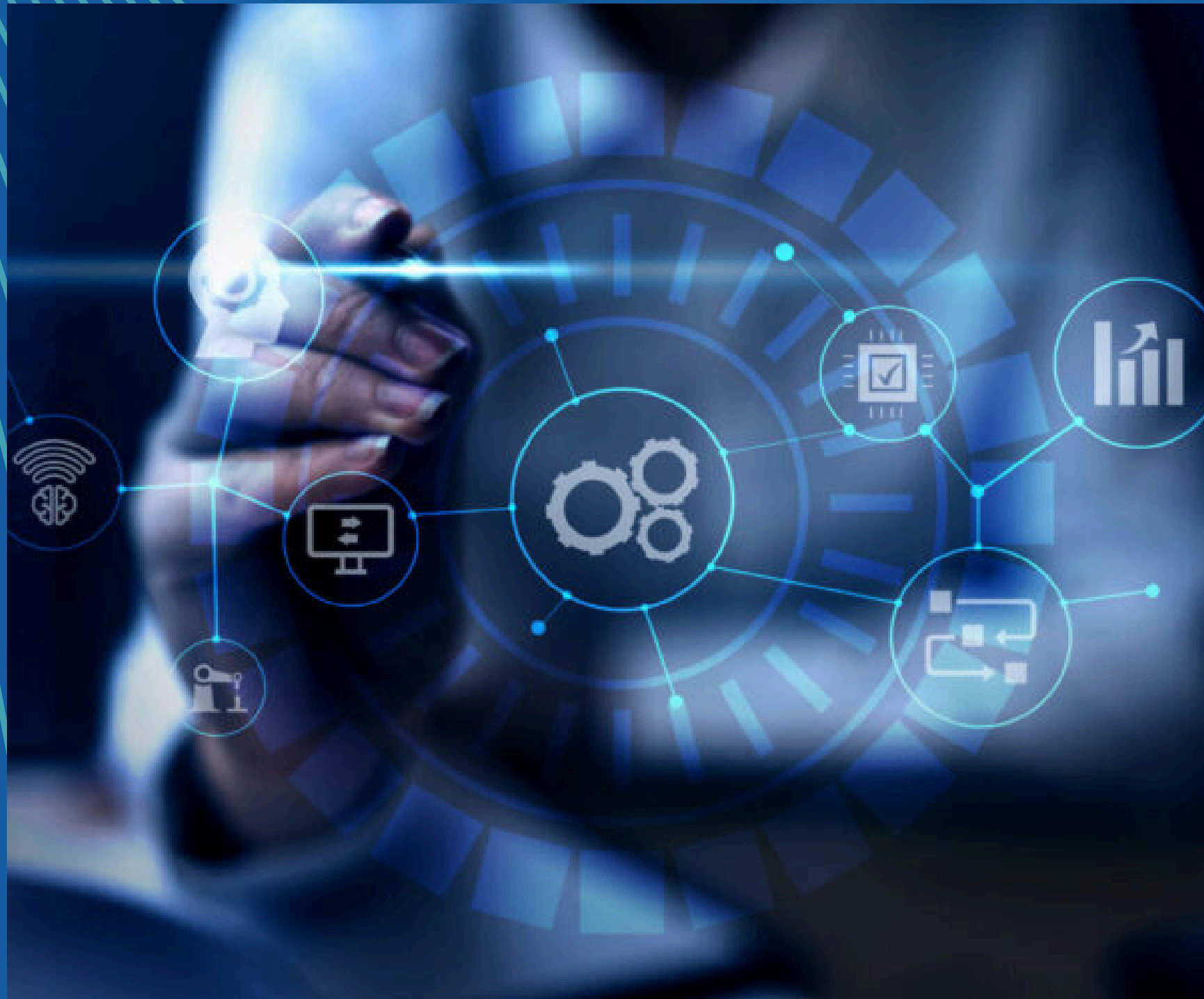
- A high-accuracy, low-latency API built with FastAPI.
- Capable of processing live video/audio streams with a target performance of >15 FPS on a standard GPU.
- Implements multi-model fusion (CNN + RNN) for robust detection of presentation attacks.

Multilingual Authentication Capability:

- A trained deep learning model specifically tuned for Urdu language acoustic features (e.g., Mel-frequency cepstral coefficients - MFCCs, prosody).
- A novel, curated dataset of Urdu spoken phrases, including genuine and synthetic (TTS) samples, to be published as a research contribution.

Forensic Source Identification Module:

- A Python-based module for post-attack analysis.
- Extracts and analyzes EXIF metadata, sensor pattern noise (SPN), and codec artifacts to attribute media to a source device or generative model.





Expected Outcomes

Deliverables:



Web-based Monitoring Interface:

An interactive dashboard built with Streamlit or React.

Provides real-time visualization of authentication attempts, system confidence scores, and forensic logs.

Features a searchable database of incidents for security auditing.

Comprehensive Documentation:

Technical Report: Full architecture, algorithm selection rationale, and performance benchmarks.

API Documentation: Clear endpoints and payload structures for developers (using OpenAPI/Swagger).

Deployment Guide: Step-by-step instructions for containerized deployment via Docker.

Impact:

Direct Application: Provides a plug-and-play security solution for sectors like digital banking (e.g., secure customer onboarding) and remote legal proceedings (e.g., verifying witness identity).

Research Contribution: Advances the field of deepfake detection in low-resource languages and integrates digital forensics directly into the authentication pipeline.



THANKYOU!

THIS WAS ALL FROM OUR SIDE